

Carbon Neutral Housing



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Table of Contents

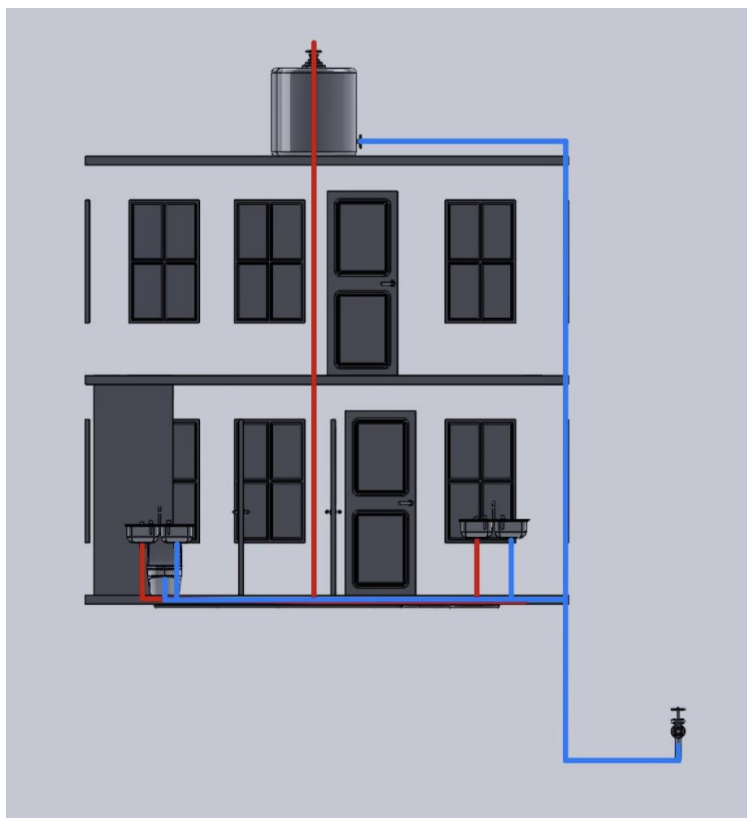
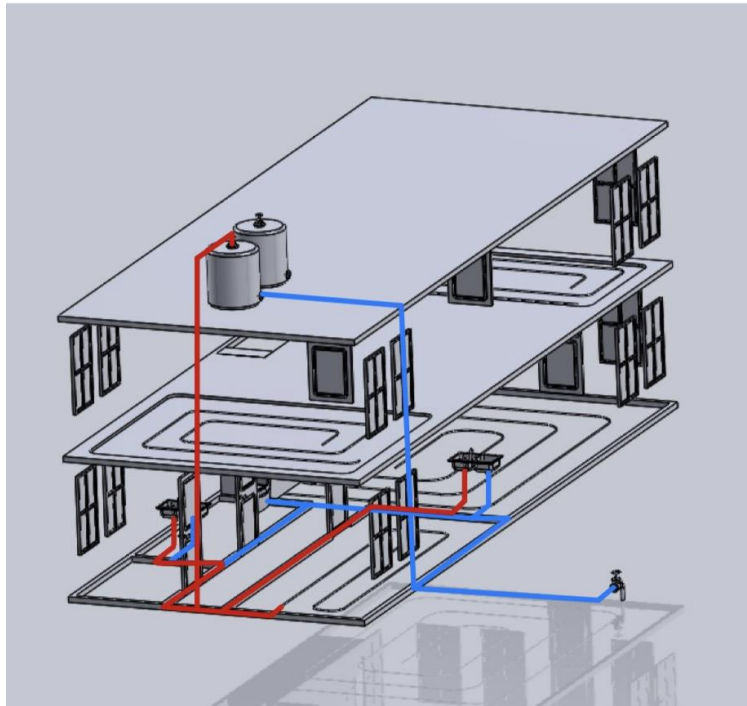
1. Proposal summary	3
2. Sketch views of the concept of the interventions installed	4
3. Intervention 1: Heat transfer	4
4. Intervention 2: Pipeline design	7
5. Intervention 3: Lighting	10
6. Intervention 4: Solar Panels	11
7. References.....	13

1. Proposal summary

This report summarises possible interventions that can be implemented to improve the energy performance of a simple three-bedroom house. To make the house carbon neutral, four main areas have been considered and analysed:

1. Heating: Good and efficient insulation is key to reduce the energy required to maintain a constant comfortable temperature in the house. For the walls polyurethane was suggested as an insulating material, for the floors, plywood was proposed and for the roof glass wood was analysed. Additionally, it was suggested that the windows be triple glazed. These adjustments were able to lower the heat transfer 66% compared to the original house.
2. Piping network: An efficient piping network was proposed for both the hot and cold-water pipes which supply water to the appliances in the house. The piping network was designed to minimise the length of the pipes and number of pipe fittings to reduce the energy consumption. The piping network required a pump with a power of around 210W.
3. Lighting: The lighting was considered as an intervention to reduce the energy use. Replacing the existent incandescent lights with LEDs was suggested as this led to an 86% reduction in power consumption for lighting.
4. Solar panels: To make the house carbon neutral, the house had to be powered by renewable resources and not fossil fuels. Therefore, installing solar panels on the house's south facing roof was suggested to be able to power the house off-grid.

2. Sketch views of the concept of the interventions installed



3. Intervention 1: Heat transfer

For the first intervention, the effectiveness of adding insulation to the walls was investigated by comparing the heat loss produced by an insulating and uninsulated wall.

The general Heat Equation:

$$Q = qA = \frac{(T_{inside} - T_{outside})A}{\left(\frac{1}{h_{inside}}\right) + \left(\frac{W_{layer1}}{k_{layer1}}\right) + \left(\frac{W_{layer2}}{k_{layer2}}\right) + \left(\frac{W_{layer3}}{k_{layer3}}\right) + \dots + \left(\frac{W_N}{k_N}\right) + \left(\frac{1}{h_{outside}}\right)}$$

$$U = \frac{1}{\frac{1}{h_{inside}} + \frac{W_{layer1}}{k_{layer1}} + \frac{1}{h_{outside}}}$$

Where the Variables are:

Q	General Heat Equation	
q	Heat flux	
A	Surface Area	202.44 m ²
T_{inside}	Temperature inside	25°C
$T_{outside}$	Temperature outside	10°C
h_{inside}	Inside convection coefficient	3.0 W/m ² k
$h_{outside}$	Outside convection coefficient	14.0 W/m ² k
W_{layer1}	Thickness of material layer N	
k_{layer1}	Thermal conductivity of material N	

Walls:

Inappropriate wall insulation accounts for 35% of heat loss in the home, so properly insulating the walls will have the largest impact in increasing energy efficiency. (1)

The current wall produced a U-value of 1.373 and a total heat flow Q of 4170W.

The insulating materials analysed were assumed to have a consistent thickness of 0.1m to evaluate uniquely the material performance.

Insulating Material	Thickness (W)(m)	Thermal conductivity (k)(W/mk)	U value (W/m ² k)	Q value (W)
Air	0.1	0.025	0.277	689.39
Glass Fibre Quilt	0.1	0.04	0.344	1045.39
Mineral Wool	0.1	0.038	0.329	1000.09
Polyurethane foam	0.1	0.025	0.227	689.39
Cellulose	0.1	0.04	0.344	1045.39
Fiberglass	0.1	0.25	1.243	3773.29

Although air is a good insulator, indicated by its low U-value (0.277), cavities are not always practical and provide little acoustic insulation. Other than air, polyurethane foam was found to have an equally low U-value, with a small total heat flow of 689.39W.

Comparing the new Q value to original:

$$\text{Percentage difference} = \frac{4170 - 689.39}{4170} \cdot 100 = 83.47\%$$

A reduction of nearly 84% in heat loss was achieved by adding only 0.1m of insulation. So, if the thickness was increased or more insulating materials layers were added, the heat loss could be reduced even more.

Roof:

The roof counts for 20% of the heat loss and is therefore the second largest contributor. The current roof produced a U-value of 3.34 and a total heat flow, Q of 5102.91W

Insulating material	Thickness (W) (m)	Thermal conductivity (k) (W/mK)	U value (W/m ² K)	Q value (W)
Glass wool	0.3	0.03	0.096	291.85
Rock wool	0.2	0.03	0.1414	347.51
Cellulose	0.2	0.04	0.185	429.42
Roofing tile	0.01	0.84	2.4	7287.84
Glass fiber quilt	0.1	0.04	0.34	1045.39
Plasterboard	0.01	0.16	2.14	185.10

Floor:

Although the floors are not the most essential component of heat loss, they still account for a 10%. The current floor produced a U-value of 0.933 and a Q value of 1613.92

Insulating material	Thickness (W) (m)	Thermal conductivity (k) (W/mK)	U value (W/m ² K)	Q value (W)
Concrete	0.1	1.4	1.355	2342.79
Plasterboard	0.1	0.16	0.971	2948.84
Plywood	0.1	0.14	0.894	2713.56

Windows:

Assuming single plane glass window were used, they produced a U value of 2.436 and a Q value of 329.98

	Materials	Thickness (W) (m)	Thermal conductivity (k) (W/mK)	U-value (W/m ² K)	Q value (W)
Triple Glazed	Glass	0.006	1.05	0.537	72.748
	Air	0.018	0.025		

Q before: 4170 + 5102.91 + 1613.92 + 329.98 = 11216.81

Q after: 689.39 + 291.85 + 2713.56 + 72.748 = 3767.55

$$\text{Percentage difference} = \frac{11216.81 - 3767.55}{11216.81} \cdot 100 = 66.41\%$$

By adding insulation, the total Q value for the house was reduced from 11216.81W to 3767.55W, which is a heat flow percentage reduction of 66.41%.

4. Intervention 2: Pipeline design

To determine the total pressure loss in the pipe network, the pressure drop due to friction, elevation and the pipe fittings were calculated.

Variables:

ΔP_f	Pressure drops due to friction (Pa)	
ΔP_c	Pressure drop due to pipe fittings (Pa)	
Q	Volumetric flow rate (m ³ /s)	0.0008
A	Cross-sectional are (m ²)	202.44
u	Velocity (m/s ²)	
Re_d	Reynolds number	
f	Friction factor	
C_f	Skin friction factor	
p	Pressure (Pa)	
h	Head (m)	
ρ	Density (kg/m ³)	1000
μ	Viscosity (Ns/m ²)	0.00089
d	Pipe diameter	0.05
e	Roughness values of pipes	
L	Length of piping	

$$\text{Pipe area, } A = \pi \left(\frac{d}{2} \right)^2 = \pi \left(\frac{0.05}{2} \right)^2 = 0.0019363 = 1.96 \cdot 10^{-3}$$

$$u = \frac{Q}{A} = \frac{0.0008}{1.96 \cdot 10^{-3}} = 0.40816 = 0.41$$

$$Re_d = \frac{\rho u d}{\mu} = \frac{1000 \cdot 0.41 \cdot 0.05}{0.00089} = 23033.70787$$

$$\text{relative roughness, } R = \frac{e}{d} = \frac{0.0015}{50} = 0.0003$$

Using Colebrook equation, find:

$$f^{-0.5} = -2 \log \left(0.27R + \frac{2.52}{Re_d \sqrt{f}} \right)$$

$$f = 0.0251411$$

Pressure loss due to friction:

Using the Darcy-Weisbach equation the pressure drop due to fiction is calculated:

$$P_f = \frac{fL}{d} \left(\frac{1}{2} \rho u^2 \right)$$

Hot water piping:

$$L_{hot} = 128.3 \text{ (m)}$$

$$P_{f \text{ hot}} = \frac{0.0251411 \times 128.5}{0.05} \left(\frac{1}{2} \cdot 1000 \cdot 0.41^2 \right) = 5422.001$$

Cold water piping:

$$L_{\text{cold}} = 28.6 \text{ (m)}$$

$$P_{f \text{ cold}} = \frac{0.0251411 \times 28.6}{0.05} \left(\frac{1}{2} \cdot 1000 \cdot 0.41^2 \right) = 1208.69$$

Total pressure lost:

$$P_f = P_{f \text{ hot}} + P_{f \text{ cold}} = 5422.00 + 1355.43 = 6630.69$$

Pressure loss due to pipe fittings:

$$P_c = C \left(\frac{1}{2} \rho u^2 \right)$$

Loss coefficients:

$$\text{T-junction, } C=1.8: P_{CT} = 1.8 \left(\frac{1}{2} \cdot 1000 \cdot 0.21^2 \right) = 151.29$$

$$90^\circ \text{ elbow, } C=0.9: P_{C \text{ elbow}} = 0.9 \left(\frac{1}{2} \cdot 1000 \cdot 0.21^2 \right) = 75.645$$

Total pressure loss due to pipe fittings: $\cdot T \text{ fittings} + \cdot \text{Elbows}$

$$P_C = (4 \cdot 151.29) + (32 \cdot 75.645) = 3025.8 \text{ (Pa)}$$

Pressure loss due to elevation:

$$A_p = \rho g h$$

$$h = 7.08 \text{ (m)}$$

$$A_p = 1000 \cdot 9.81 \cdot 7.08 = 69454.8 \text{ (Pa)}$$

Head loss:

$$\text{Head loss from friction: } \Delta H_f = \frac{P_C}{\rho g}$$

$$\Delta H_f = \frac{3025.8}{1000 \cdot 9.81} = 0.30844$$

$$\text{Head loss from pipe fittings: } \Delta H_f = C \left(\frac{1}{2} \cdot \frac{u^2}{g} \right)$$

$$\Delta H_{fT} = 1.8 \left(\frac{1}{2} \cdot \frac{0.41^2}{9.81} \right) = 0.007711$$

$$\Delta H_{f \text{ elbow}} = 0.9 \left(\frac{1}{2} \cdot \frac{0.41^2}{9.81} \right) = 0.003427$$

$$\text{Total head loss: } \Delta H_f + \Delta H_{fT} + \Delta H_{f \text{ elbow}}$$

$$\Delta P_h = 0.3084 + (4)0.007711 + (32)0.0033427 = 0.44625$$

Total pressure loss:

$$\Delta P = \Delta P_h + \Delta P_c + A_p$$

$$\Delta P = 6630.69 + 0.44625 + 3025.8 + 69454.8 = 79111.74$$

Water pump:

A water pump is required at the bottom of the house, next to the water outlet. Assuming 60% efficiency and a safety factor of 2 the power loss is calculated.

$$\text{Theoretical power required, } P = Q \cdot \Delta P = 0.008 \cdot 79111.74 = 63.289 \text{ (W)}$$

$$\text{Power rating of pump} = \frac{63.289}{0.6} \cdot 2 = 210.964 \text{ (W)}$$

5. Intervention 3: Lighting

LED lights are an easy way to reduce energy consumption within the home. Additionally, they last longer and are more user-friendly as they don't have to be changed as often.

The reduction in energy use is calculated to analyse the impact of using LEDs across the house.

The selected LEDs produce 1521 lumens per bulb at 13 W (1)

$$\text{lumens required} = \text{area} \cdot \text{lux}$$

$$\text{bulbs required} = \frac{\text{lumens required}}{\text{lumens per bulb}}$$

Assuming the lights are used for 7 hours per day:

$$\text{Energy for bulbs} = \frac{\text{Power} \cdot \text{Time used}}{1000}$$

$$E = \frac{13 \cdot 7}{1000} = 0.091 \text{ kwh}$$

Room	Area	Lux	Lumens	Number of bulbs	Energy of LED	Energy of non-LEDs
Kitchen	20	300	6000	4	0.364	2.8
Bedroom 1	18.84	150	2826	2	0.182	1.4
Bedroom 2	9.84	150	1476	1	0.092	0.7
Bedroom 3	10.20	150	1530	1	0.092	0.7
Bathroom	3.68	300	1104	1	0.092	0.7
Living room	15.85	200	3170	2	0.182	1.4
Office	9.95	150	995	1	0.092	0.7
Total Energy					1.096	8.4

Comparing the incandescent lightbulbs producing 1521 lumens require 100W:

$$\text{Energy for bulbs} = \frac{100 \cdot 7}{1000} = 0.7 \text{ kwh}$$

$$\text{Total Energy} = 8.4 \text{ kwh}$$

$$\text{Percentage difference} = \frac{8.4 - 1.096}{8.4} \cdot 100 = 86.95\%$$

Replacing the incandescent lightbulbs with LED ones would reduce energy use by 86.95%.

6. Intervention 4: Solar Panels

The house will use energy, but for it to be carbon neutral, the energy should be renewable sourced and not come from fossil fuels. For the fourth intervention, the house was suggested to be powered by solar panels installed on the roof.

Factors affecting amount of energy produced by solar panels:

1. Size: The amount of energy generated is proportional to its total area. So, to maximise the energy generated, the space must be used at its full potential.
2. Direction: Due to the path the sun takes, the solar panels will get more direct sunlight if they're placed facing the South and at an angle of 35 degrees. (2)
3. Type: The quality of material used has a significant effect on the power generated
4. Amount of sunlight: Depending on the time of the year the available daylight varies significantly. During the winter, when days are shorter, and sunlight is scarce, the solar panels will generate less energy compared to summer, when days are longer, and sunlight is stronger.

Available space:

The area of the roof is 35.41 m² and it has an incline of 50 degrees which is close to the optimal 35 degrees.

The standard solar panel size is a 2m length x 1m width rectangle (2 m²) (3)

$$\text{Maximum number of panels} = \frac{35.41}{2} = 17.7 = 17 \text{ solar panels}$$

Daily output for all panels:

The average solar panels outputs between 250-400W (4)

On average there's 4.3 hours of daylight in the UK. (5)

$$\text{Total energy output for the 17 panels} = 17(325) = 5525 \text{ (kW/y)}$$

Energy consumed by household appliances:

Appliance	Power	Quantity	Running time	Energy	Energy
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	consumption (W)		(h)	consumption per day (Wh/d)	consumption per year (Wh/y)
Electric kettle	1200	1	0.1	120	43800
Fridge	400	1	24	9600	3504000
Toaster	1800	1	0.1	180	65700
Oven	2150	1	0.5	1075	392375
Microwave	600	1	0.3	180	65700
TV	60	1	2	120	43800
Washing machine	500	1	2	1000	365000
Heating	693	1	5	3767	1374955
Water pump	210.964	1	2	421.93	154004.19
Lighting	221	1	7	1547	564655
Total	7834.964			1810	6573650

The data for the appliances was sourced from DaftLogic (5) and this reports calculations.

While the suggested solar panel solution could meet a substantial portion of the energy demand, it would not be sufficient to completely power the off-grid house as the total energy consumption outweighs the energy produced: 6573kW/y > 5525 kW/y.

To solve this, more solar panels could be installed on the other side of the roof, another renewable energy source, like wind energy, could be added to obtain the necessary energy to power the house or more investigation could be done to reduce the energy consumed, especially in the heating intervention which is still quite high.

7. References

- (1) <https://www.northernenergy.co.uk/news/the-4-main-sources-of-heat-loss-in-the-home/>
- (2) <https://www.geogreenpower.com/blog/a-guide-for-where-to-place-your-solar-panels/#:~:text=With%20a%20south%2Dfacing%20roof,receive%20very%20little%20direct%20sunlight.>
- (3) <https://www.theecoexperts.co.uk/solar-panels/how-many-do-i-need#one>
- (4) [https://www.forbes.com/home-improvement/solar/how-much-power-does-a-solar-panel-produce/#:~:text=Average%20Solar%20Panel%20Production,-The%20average%20solar&text=On%20average%2C%20a%20standard%20residential,\(kWh\)%20of%20electricity%20annually.](https://www.forbes.com/home-improvement/solar/how-much-power-does-a-solar-panel-produce/#:~:text=Average%20Solar%20Panel%20Production,-The%20average%20solar&text=On%20average%2C%20a%20standard%20residential,(kWh)%20of%20electricity%20annually.)
- (5) <https://www.statista.com/statistics/322575/average-daily-sun-hours-in-the-united-kingdom-uk/#:~:text=Average%20daily%20sun%20hours%20in%20the%20United%20Kingdom%202001%2D2023&text=Since%20the%20turn%20of%20the,daily%20sun%20hours%20was%20recorded.>